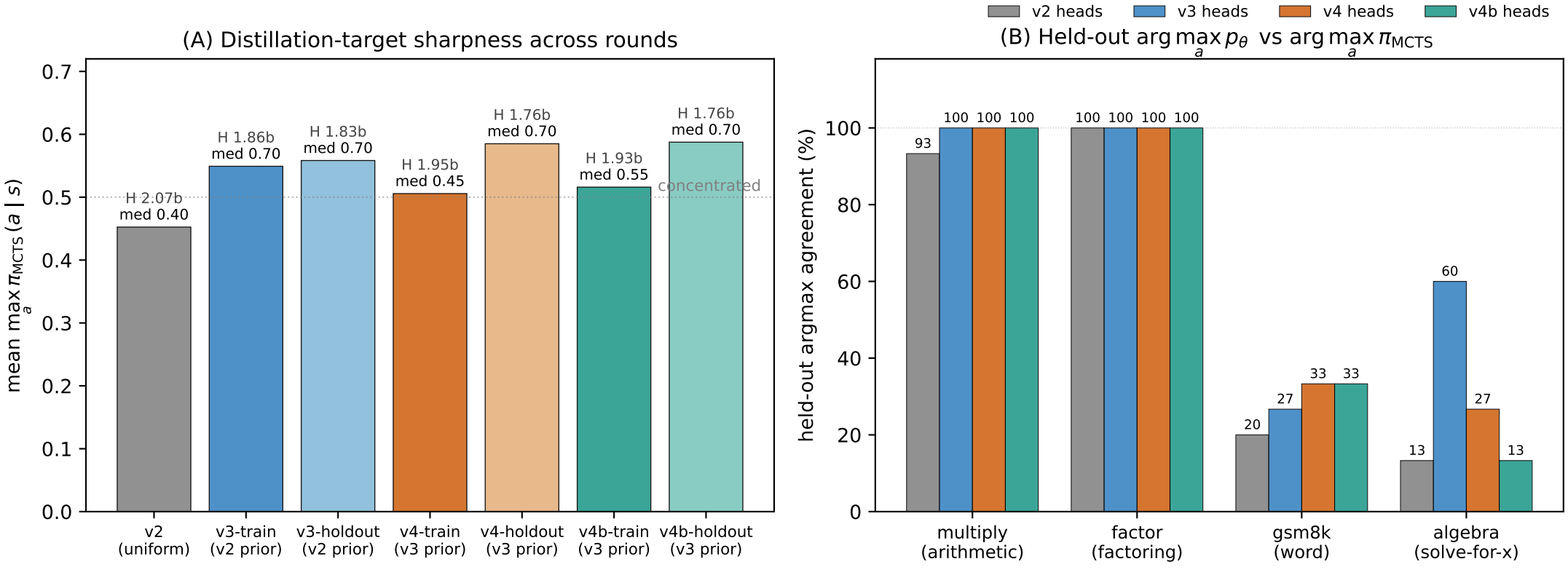
(A) Distillation-target sharpness across rounds
mean $\max_a \pi_{\text{MCTS}}(a \mid s)$
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
H 2.07b
med 0.40
H 1.86b
med 0.70
H 1.83b
med 0.70
H 1.95b
med 0.45
H 1.76b
med 0.70
H 1.93b
med 0.55
H 1.76b
med 0.70
concentrated
v2 (uniform)
v3-train (v2 prior)
v3-holdout (v2 prior)
v4-train (v3 prior)
v4-holdout (v3 prior)
v4b-train (v3 prior)
v4b-holdout (v3 prior)
v2 heads
v3 heads
v4 heads
v4b heads
(B) Held-out $\arg\max_a p_\theta$ vs $\arg\max_a \pi_{\text{MCTS}}$
held-out argmax agreement (%)
0
20
40
60
80
100
93
100
100
100
100
100
100
100
20
27
33
33
13
60
27
13
multiply (arithmetic)
factor (factoring)
gsm8k (word)
algebra (solve-for-x)